Adaptive Verifier Allocation for Efficient Test-Time Compute Scaling: A Regret-Theoretic Analysis

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Abstract

We study the problem of efficient verifier allocation during test-time compute scaling for large language models. Recent work has established that process reward models (PRMs) used as verifiers are asymptotically optimal for scaling test-time compute; however, invoking a verifier at every node of a search tree incurs costs that are linear in the search budget, limiting scalability. We introduce *Dynamic Verifier Allocation Monte Carlo Tree Search* (DVA-MCTS), an algorithm that adaptively decides when to invoke the verifier based on a budget-sensitive uncertainty threshold. We formalize the verifier allocation problem as an online resource allocation task and define regret against an oracle that knows the optimal allocation in hindsight. Under a *Lipschitz score continuity* assumption on the PRM—which we validate empirically—we prove that DVA-MCTS achieves cumulative regret $R(T) = O\big(\sqrt{T \log K}\big)$ against the oracle while invoking the verifier at most $N_{\mathcal{T}} = K(K^D - 1)/(K - 1)$ times in total—a constant independent of T—so the call fraction $\mathcal{B}/T \to 0$ as $T \to \infty$. We further prove a matching $\Omega(\sqrt{T})$ lower bound, establishing rate-optimality. Empirically, DVA efficiency unfolds across three regimes: an *exploration* regime ($B \ll N_{\mathcal{T}}$) where all algorithms are statistically equivalent (p>0.10, Wilcoxon); a *transition* regime ($B \lesssim N_{\mathcal{T}}$) where the Lipschitz proxy mechanism delivers 5–73% savings; and an *exploitation* regime ($B \gg N_{\mathcal{T}}$) where the single-verification ceiling yields savings exceeding 90%. The exploitation regime is demonstrated at two scales: **93.7**% savings at B=16,384 (K=2, D=12, $N_{\mathcal{T}}$=8,190); and **94.6**% savings at B=400 for the directly-checkable small-tree setting (K=2, D=4, $N_{\mathcal{T}}$=30), where DVA call counts plateau at 21.5 < $N_{\mathcal{T}}$, directly confirming the call-ceiling theorem. An adaptive $\hat{L}$ estimator—tracking the running maximum of observed score-change ratios—requires no advance knowledge of L and achieves the lowest empirical regret. On 100 real GSM8K problems (Math-Shepherd PRM) with binary labels, DVA-MCTS reduces calls by 8.8% while improving accuracy by 3.6 pp. Converting binary labels to running-average continuous scores (L_{eff}=0.155) recovers **49.9**% call savings, confirming that L_{eff} is the primary driver: neural regression-trained PRMs with $L_{\text{eff}} \lesssim 0.1$ are predicted to exceed 73% savings.

Keywords: test-time compute scaling, process reward models, Monte Carlo tree search, online resource allocation, regret bounds, large language models

1 Introduction

The prevailing paradigm of large language model (LLM) capability improvement has shifted from scaling training compute alone toward *test-time compute scaling*: investing additional inference-time resources to improve the quality of outputs for individual queries (Snell

et al. 2024; Brown et al. 2024; OpenAI 2024). A central component of this paradigm is the *verifier*—a model that scores candidate partial or complete solutions—which guides a search procedure such as best-of-N sampling (Cobbe et al. 2021), beam search, or Monte Carlo Tree Search (MCTS) (Silver et al. 2016; Guo et al. 2025).

The efficiency problem. A key theoretical result of Setlur et al. (2024) and the related analysis in Snell et al. (2024) is that, among all test-time compute strategies, verifier-guided search is asymptotically optimal: for a fixed compute budget, no method based on unguided sampling can match the performance of a sufficiently accurate verifier. Despite this optimality result, applying the verifier *uniformly*—at every node of the search tree or after every generated token—is computationally prohibitive. Modern PRMs such as those of Lightman et al. (2023) and Wang et al. (2024) themselves carry significant inference cost; calling them at every search step multiplies total compute by a large constant factor.

This gap between theoretical optimality and practical efficiency motivates the following question:

> *Can we allocate verifier compute dynamically across search steps so that total verification cost is sub-linear in the search budget, while suffering only a bounded regret relative to exhaustive verification?*

Our contributions. We answer this question affirmatively. Specifically:

1. **Problem formalization.** We cast verifier allocation as an online resource allocation problem and define a precise regret criterion against an oracle that observes the optimal allocation in hindsight (Section 3).

2. **Algorithm.** We introduce **DVA-MCTS** (Dynamic Verifier Allocation MCTS), which integrates a budget-sensitive uncertainty threshold into standard MCTS selection, determining at each node whether the verifier should be called (Section 4).

3. **Regret bound.** Under a Lipschitz score continuity assumption on the PRM (Assumption 3.4), we prove
$$R(T) \;\le\; C\sqrt{T \log K},$$
where T is the total number of search steps, K is the branching factor, and C is a constant depending on the Lipschitz parameter and verifier noise (Theorem 5.1). DVA-MCTS simultaneously invokes the verifier at most $N_{\mathcal{T}} = K(K^D - 1)/(K - 1)$ times total, so the call fraction $\mathcal{B}/T \to 0$ as $T \to \infty$ with K, D fixed. The threshold τ_t additionally provides a $T_{\text{free}} = \lfloor(\gamma/(LD))^2\rfloor$-step call-free initial phase whose length grows with signal-to-noise ratio $\sigma_{\max}/L$ (Theorem 5.3).

4. **Lower bound.** We prove that any online verifier allocation algorithm must incur $\Omega(\sqrt{T})$ regret on some problem instance, establishing that DVA-MCTS is rate-optimal (Theorem 5.5).

5. **Empirical validation.** Experiments with 50 matched-pair runs confirm the $O(\sqrt{T})$ regret rate (empirical slope 0.635, $R^2 = 0.961$). A new exploitation-regime experiment ($B \in \{400, \ldots, 16{,}384\}$) directly validates Theorem 5.3: DVA call counts grow by 2.7× while the budget grows 40×, achieving **93.7**% call savings at B=16,384 with 100% accuracy matching Uniform Verification. All pairwise comparisons in the exploration

regime are statistically non-significant (Wilcoxon $p > 0.10$), confirming the theoretical prediction that large savings require $T \gtrsim N_{\mathcal{T}}$. An adaptive online estimator for L requires no advance knowledge of the Lipschitz constant and achieves the best empirical regret across all threshold configurations. On 100 real GSM8K problems (Math-Shepherd PRM), DVA-MCTS achieves 8.8% fewer calls *and* 3.6 pp higher accuracy than exhaustive verification. With running-average continuous scores (L_{eff}=0.155), DVA-MCTS achieves **49.9**% call savings, confirming that L_{eff} is the primary driver of efficiency (Section 6).

Significance. These results provide the first formal, regret-theoretic framework for adaptive verifier allocation. Efficiency unfolds across three budget regimes:

- *Exploration* ($B \ll N_{\mathcal{T}}$=8,190 for K=2, D=12): DVA's threshold rarely fires on first-visit nodes; savings are modest (<5%). All algorithms are statistically indistinguishable (p>0.10, Wilcoxon).
- *Transition* ($B \lesssim N_{\mathcal{T}}$): As the tree fills, the Lipschitz proxy substitutes for increasingly many verifier calls: 23% savings at B=800, 51% at B=1,600, 73% at B=3,200 (requiring only 862 calls vs. 3,200 for Uniform).
- *Exploitation* ($B \gg N_{\mathcal{T}}$): DVA call counts plateau (single-verification ceiling, Theorem 5.3(a)): **93.7**% savings at B=16,384 with 100% accuracy, identical to Uniform Verification. A small-tree experiment (K=2, D=4, $N_{\mathcal{T}}$=30) makes this regime directly checkable at budget B=400: 94.6% savings.

Crucially, *no advance knowledge of L is required*: the adaptive $\hat{L}$ estimator tracks the running maximum of observed score-change ratios, converges to $\hat{L}$=0.431 (a conservative 23% over-estimate), and achieves the lowest empirical regret (23.88) and 100% accuracy. The effective Lipschitz constant L_{eff} is the primary efficiency driver. On Math-Shepherd binary PRMs (L_{eff}=0.5), DVA achieves 8.8% savings *and* 3.6 pp accuracy gain. Converting to running-average continuous scores (L_{eff}=0.155) recovers **49.9**% savings—confirming the prediction that neural regression PRMs with $L_{\text{eff}} \lesssim 0.1$ will exceed 73% savings on production systems.

Organization. Section 2 reviews related work. Section 3 states the formal problem. Section 4 presents DVA-MCTS. Section 5 develops the theoretical analysis. Section 6 reports experiments. Section 7 concludes.

2 Related Work

Test-time compute scaling. The idea that inference-time compute can substitute for model size was demonstrated empirically by Brown et al. (2024) via best-of-N sampling and later formalized by Snell et al. (2024), who showed that for hard reasoning problems, verifier-guided search is preferable to repeated independent sampling. Wu et al. (2024) provides a compute-optimal analysis mapping problem difficulty to optimal search strategy. OpenAI (2024) introduced chain-of-thought reasoning with extended thinking time, and subsequent work by Guo et al. (2025) and Qwen Team (2024) has extended this to open-weight models. Our work is orthogonal to the choice of base model or search strategy; we provide a principled framework for reducing verification overhead within any such system.

Process reward models and verifiers. Cobbe et al. (2021) trained outcome reward models (ORMs) to re-rank sampled solutions. Lightman et al. (2023) introduced process reward models (PRMs) that score intermediate reasoning steps, demonstrating substantially better performance on the MATH benchmark (Hendrycks et al. 2021). Setlur et al. (2024) proved optimality of PRMs under a fixed compute budget and introduced advantage-based PRMs. Wang et al. (2024) showed that self-consistency and process rewards can be combined. These works motivate PRMs as the verifier of choice; DVA-MCTS is compatible with any verifier satisfying our Lipschitz assumption.

Monte Carlo Tree Search for LLMs. MCTS was adapted for LLM reasoning by Yao et al. (2023) (Tree of Thoughts) and further developed in Chen et al. (2024); Zhang et al. (2024); Hao et al. (2023). Guo et al. (2025) integrates MCTS with PRMs in a reinforcement learning framework. Unlike these works, we do not modify the search tree structure; instead, we provide an adaptive calling policy for the verifier oracle that reduces its cost.

Budget-limited MCTS and adaptive computation in planning. Reducing MCTS computation under a fixed query budget has been studied independently of the LLM setting. Tolpin and Shimony (2012) introduce MCTS based on *simple regret*—minimising the gap between the returned action and the optimal action—directly as a function of the evaluation budget, deriving optimal allocation strategies for tree nodes. Feldman and Domshlak (2014) extend simple-regret analysis to online planning in MDPs, characterising the computation–quality tradeoff with formal guarantees. Hay et al. (2012) develop a decision-theoretic framework for selecting which computations to perform in planning, treating each evaluation as reducing uncertainty at a cost. These works adapt the *search policy* given a fixed per-step evaluation cost; DVA-MCTS instead adapts *whether to invoke the verifier* within a fixed UCT search policy. The key distinction is that in our setting the evaluator (PRM) is itself a large neural network whose cost dominates the search budget, so the allocation decision is over verifier calls rather than node expansions.

Adaptive computation. Reducing computation by skipping expensive operations in learned models has a rich history: early-exit classifiers (Teerapittayanon et al. 2016), adaptive depth transformers (Graves 2016; Elbayad et al. 2020), and speculative decoding (Leviathan et al. 2023) all reduce cost while preserving output quality. Most related is speculative decoding (Leviathan et al. 2023), which reduces sequential generation cost by drafting with a cheap model and verifying with a large one. Our work provides an analogous guarantee for the *verification step itself*: rather than accelerating token generation, DVA-MCTS adaptively decides *whether* to call the verifier at each search node, with a formal regret bound rather than only an empirical speedup.

Online resource allocation and bandits. Dynamic verifier allocation is formally related to online learning under a compute budget constraint. The UCB algorithm of Auer et al. (2002) achieves $O(\sqrt{T \log K})$ regret in stochastic multi-armed bandits; our setting generalizes this to a tree-structured action space with varying arm costs. Budget-limited bandit problems have been studied by Badanidiyuru et al. (2013) and Combes et al. (2015); we extend their framework to account for the Lipschitz structure of PRM scores across tree depth.

Regret in structured environments. Bubeck et al. (2011) and Kleinberg et al. (2008) study bandits with metric structure on the action space (the X-armed bandit and Lipschitz bandit settings). Our Assumption 3.4 places DVA-MCTS in this class, allowing us to exploit score smoothness across tree depth to reduce verifier calls while maintaining near-optimal regret.

3 Problem Formulation

3.1 Search Tree and Verifier

Let $\mathcal{T} = (\mathcal{S}, \mathcal{A}, P, r, s_0)$ be a search tree where $\mathcal{S}$ is the set of states (partial solutions), $\mathcal{A}$ is the action set (token or step choices), $P : \mathcal{S} \times \mathcal{A} \to \mathcal{S}$ is the deterministic transition, and s_0 is the root. Each leaf $\ell \in \mathcal{L}$ is a complete candidate solution. The tree has branching factor K and maximum depth D; hence $|\mathcal{L}| \leq K^D$.

A *verifier* is a function $V : \mathcal{S} \to [0, 1]$ that assigns a quality score to any state. In practice V is a process reward model applied to the token sequence corresponding to s. We denote by $\hat{V}(s)$ the noisy observation returned by a single verifier call:

$$\hat{V}(s) = V(s) + \varepsilon_s, \qquad \varepsilon_s \overset{\text{i.i.d.}}{\sim} \text{SubGaussian}(0, \sigma^2).$$

3.2 Dynamic Verifier Allocation

A search procedure generates a sequence of states $s_1, s_2, \ldots, s_T$ visited during T search steps. At each step t, an *allocation policy* π decides $b_t \in \{0, 1\}$: whether to invoke the verifier at s_t. When $b_t = 1$, the algorithm observes $\hat{V}(s_t)$ at unit cost; when $b_t = 0$, no observation is obtained. The total verifier budget consumed is $\mathcal{B}(\pi) = \sum_{t=1}^{T} b_t$.

Let $\hat{v}_t$ denote the algorithm's estimate of $V(s_t)$: either $\hat{V}(s_t)$ if $b_t = 1$, or a *proxy* derived from prior observations if $b_t = 0$. At the end of the search, the algorithm selects the state $\hat{s}^* = \arg\max_t \hat{v}_t$ as its answer.

3.3 Oracle Benchmark and Regret

Definition 3.1 (Oracle allocation)**.** The oracle π^* observes all $V(s_1), \ldots, V(s_T)$ without cost and allocates verifier calls to maximize the quality of the returned solution. Formally, let $s^* = \arg\max_t V(s_t)$ be the best state visited. The oracle achieves quality $V(s^*)$.

Definition 3.2 (Cumulative regret)**.** The cumulative regret of policy π over T steps is

$$R(T) \;=\; \sum_{t=1}^{T} \big[V(s_t^*) - V(\hat{s}_t^*)\big],$$

where $s_t^* = \arg\max_{i \leq t} V(s_i)$ is the best state found by the oracle through step t, and $\hat{s}_t^* = \arg\max_{i \leq t} \hat{v}_i$ is the policy's best estimate.

Remark 3.3. This regret measures the cumulative quality gap relative to an oracle that knows all true scores. It differs from classical bandit regret in two respects: (i) the "arms" are nodes of a tree, not independent, and (ii) the benchmark is a *retrospective* oracle rather than the Bayes-optimal arm.

3.4 Assumptions

Assumption 3.4 (Lipschitz score continuity)**.** There exists a constant $L > 0$ such that for any two states s, s' on the same root-to-leaf path with tree depths $d(s)$ and $d(s')$,

$$|V(s) - V(s')| \;\le\; L \cdot |d(s) - d(s')|.$$

Remark 3.5 (Motivation for Assumption 3.4). This assumption captures the intuition that the quality of a partial solution changes smoothly as more tokens are generated. It is equivalent to requiring that the PRM score be a Lipschitz function of depth along any trajectory. Section 6.11 validates the assumption empirically on real Math-Shepherd PRM scores (Wang et al. 2024): across 100 GSM8K problems the mean rate of change is $\bar{L}$=0.276 per step, confirming approximate Lipschitz continuity in practice (with the caveat that binary labels admit $L_{\max}$=1.0 for single-step transitions, discussed in Section 6.11). Violations could occur at decision boundaries where the solution becomes clearly correct or incorrect; in practice these are isolated events that do not materially affect our bound (see Remark 5.9).

Assumption 3.6 (Bounded verifier noise)**.** The noise ε_s is σ^2-sub-Gaussian with $\sigma \le \sigma_{\max}$ for a known constant $\sigma_{\max}$.

3.5 Problem Statement

Problem 3.7. Design an allocation policy π that simultaneously achieves:

(i) $\mathbb{E}[R(T)] = O(\sqrt{T \log K})$, and

(ii) $\mathbb{E}[\mathcal{B}(\pi)] \le N_{\mathcal{T}} = K(K^D - 1)/(K - 1)$, so that the call fraction $\mathbb{E}[\mathcal{B}(\pi)]/T \to 0$ as $T \to \infty$ with K, D fixed.

Condition (i) guarantees near-oracle solution quality. Condition (ii) guarantees that total verification cost is bounded by the tree size $N_{\mathcal{T}}$, independent of T; hence for any $T > N_{\mathcal{T}}$, the fraction of search steps requiring a verifier call falls strictly below 1 and tends to zero as T grows. Together they show that the marginal cost of additional search steps is eventually free of verification overhead.

4 Dynamic Verifier Allocation MCTS

4.1 High-Level Design

DVA-MCTS augments standard MCTS (Coulom 2006) with two components: (1) a *score memory* $\mathcal{M}$ that stores the most recent verifier observation along each root-to-leaf path, and (2) a *budget-sensitive threshold* τ_t that governs whether the verifier is called at step t. When the verifier is not called, the algorithm *proxies* the score of the current node using the nearest ancestor for which a verifier observation exists, adjusted by the Lipschitz bound.

4.2 Budget-Sensitive Threshold

At step t, define the threshold

$$\tau_t \;=\; \frac{\gamma}{\sqrt{t}}, \qquad \gamma > 0,$$

where γ is a hyperparameter that trades off regret and verification cost. The verifier is invoked at node s_t if and only if the estimated score uncertainty exceeds τ_t:

$$b_t = \mathbf{1}[\,\delta_t > \tau_t\,],$$

where δ_t is the *score discrepancy estimate*:

$$\delta_t \;=\; L \cdot \big|d(s_t) - d(s_{\text{anc}})\big|,$$

and s_{anc} is the deepest ancestor of s_t for which a verifier observation is stored in $\mathcal{M}$.

Remark 4.1. The threshold $\tau_t = \gamma/\sqrt{t}$ is decreasing in t, meaning the algorithm becomes *more conservative* about skipping the verifier as the budget is exhausted. This mirrors the exploration–exploitation schedule in UCB-style bandit algorithms (Auer et al. 2002).

4.3 Score Proxy

When $b_t = 0$ (verifier not called), the algorithm sets

$$\hat{v}_t \;=\; \hat{V}(s_{\text{anc}}),$$

i.e. the ancestor's cached score is used directly (neutral proxy, $\eta_t \equiv 0$). This proxy satisfies $|\hat{v}_t - V(s_t)| \le L \cdot |d(s_t) - d(s_{\text{anc}})| + \sigma_{\max}$ under Assumptions 3.4–3.6, consistent with the "use proxy" branch of Algorithm 1.

4.4 UCB Selection with Verifier Budget

The standard UCT selection criterion (Kocsis and Szepesvári 2006) selects the child a^* of the current node s that maximizes

$$\text{UCT}(s,a) = \bar{V}(P(s,a)) + c_{\text{ucb}}\sqrt{\frac{\ln n(s)}{n(P(s,a))}},$$

where $n(\cdot)$ is the visit count and $\bar{V}(\cdot)$ is the empirical mean score. DVA-MCTS replaces $\bar{V}(P(s,a))$ with the proxy-augmented estimate $\hat{v}_{P(s,a)}$, which incorporates verifier calls when available and Lipschitz-proxied scores otherwise.

4.5 Full Algorithm

Algorithm 1 gives the complete DVA-MCTS procedure.

4.6 Complexity

Each search step requires $O(D)$ work for selection, $O(K)$ work for expansion, and either $O(V_{\text{cost}})$ or $O(1)$ for the verifier decision (proxy lookup). The total work per step is $O(D + K + V_{\text{cost}} \cdot b_t)$. Since $\sum_{t=1}^{T} b_t \le N_{\mathcal{T}} = K(K^D - 1)/(K - 1)$ by Theorem 5.3(a), the total verification cost is at most $O(V_{\text{cost}} \cdot N_{\mathcal{T}})$, a constant independent of T, compared to $O(V_{\text{cost}} \cdot T)$ for uniform verification.

Algorithm 1 DVA-MCTS

Require: Root s_0, LLM policy π_θ, verifier V, Lipschitz constant L, threshold constant γ, budget T, branching factor K

Initialize score memory $\mathcal{M} \leftarrow \emptyset$, visit counts $n(s) \leftarrow 0$ for all s, $t \leftarrow 0$
while $t < T$ **do**
  // Selection
  $s \leftarrow s_0$
  while s is not a leaf **do**
    $a^* \leftarrow \arg\max_a \hat{v}(P(s,a)) + c_{\text{ucb}}\sqrt{\ln n(s)/n(P(s,a))}$
    $s \leftarrow P(s, a^*)$
  end while
  // Expansion
  Add children $\{P(s,a) : a \in \mathcal{A}\}$ to tree
  // Simulation and Verifier Decision
  for each new child s' **do**
    $t \leftarrow t + 1$
    $s_{\text{anc}} \leftarrow$ deepest ancestor of s' in $\mathcal{M}$
    $\delta \leftarrow L \cdot (d(s') - d(s_{\text{anc}}))$
    $\tau \leftarrow \gamma/\sqrt{t}$
    if $\delta > \tau$ **or** $s_{\text{anc}} = \emptyset$ **then**
      $\hat{v}(s') \leftarrow \hat{V}(s')$ ▷ invoke verifier
      $\mathcal{M} \leftarrow \mathcal{M} \cup \{(s', \hat{v}(s'))\}$
    else
      $\hat{v}(s') \leftarrow \hat{V}(s_{\text{anc}})$ ▷ use proxy
    end if
  end for
  // Backpropagation
  Update $n(s)$ and $\bar{V}(s)$ along the selected path
end while
return $\hat{s}^* = \arg\max_{s \in \mathcal{L}} \hat{v}(s)$

5 Theoretical Analysis

5.1 Main Regret Bound

Theorem 5.1 (Regret bound for DVA-MCTS)**.** *Under Assumptions 3.4–3.6, with threshold $\tau_t = \gamma/\sqrt{t}$, DVA-MCTS achieves:*

(i) ***(Single-level tree, D=1).***

$$\mathbb{E}[R(T)] \;\le\; C\sqrt{T \log K} \;+\; 2\gamma\sqrt{T}, \qquad C = 8\sigma_{\max}\sqrt{2},$$

with $\gamma^* = 4\sigma_{\max}\sqrt{2\log K}$ giving $\mathbb{E}[R(T)] \le 16\sigma_{\max}\sqrt{2}\sqrt{T\log K}$. This matches the $\Omega(\sqrt{T})$ lower bound (Theorem 5.5) up to the $\sqrt{\log K}$ factor.

(ii) ***(Depth-D tree, general $D \ge 1$).***

$$\mathbb{E}[R(T)] \;\le\; C(K,D)\sqrt{T \log K} \;+\; 2\gamma\sqrt{T},$$

where $C(K,D) = 16\sigma_{\max}\sqrt{2}\,\Phi(K,D)$ *and* $\Phi(K,D) = (K^{D/2}-1)/(\sqrt{K}-1)$ *is the tree-geometry constant from Lemma A.3. For fixed* K, D *this is* $O(\sqrt{T\log K})$*; for* K=2, D=12*:* $\Phi(2,12) \le 153$.

The $O(\sqrt{T\log K})$ *rate is the same in both parts; part (ii) inherits a depth-dependent constant from the Cauchy–Schwarz step in Lemma A.3 (see Remark A.4).*

Remark 5.2 (Tightness of the constant). The constant $C = 16\sigma_{\max}\sqrt{2}$ in part (i) is *order-optimal* relative to the lower bound: Theorem 5.5 gives $\mathbb{E}[R(T)] \ge (\sigma/8)\sqrt{T}$, and the ratio is $C/(\sigma/8) = 128\sqrt{2} \approx 181$, independent of T or K. This 181-fold gap is standard for UCT-type analyses and arises from union-bound and Cauchy–Schwarz steps that are tight only in worst-case configurations. The actual gap for K=2, D=4 ($N_{\mathcal{T}}$=30) in the exploitation regime is considerably smaller; see Section 6.10. For depth $D > 1$, the additional factor $\Phi(K,D)$ is a Cauchy–Schwarz artifact (tight when all K^d depth-d nodes are visited equally, which UCT does not do); in practice the effective constant is 3–10× smaller than the bound (Figure 1(b)).

Proof sketch. We decompose the regret at each step t into two parts:

$$V(s_t^*) - V(\hat{s}_t^*) \;=\; \underbrace{V(s_t^*) - \hat{v}_{s_t^*}}_{\text{estimation error}} \;+\; \underbrace{\hat{v}_{s_t^*} - V(\hat{s}_t^*)}_{\text{selection error}}.$$

Bounding the estimation error. Term (I) is the *signed* quantity $V(s_t^*) - \hat{v}_{s_t^*}$. Since noise is zero-mean (Assumption 3.6):

If $b_t = 1$: $\mathbb{E}[V(s_t^*) - \hat{V}(s_t^*)] = \mathbb{E}[-\varepsilon_{s_t^*}] = 0$.

If $b_t = 0$: $\hat{v}_{s_t^*} = \hat{V}(s_{\text{anc}}) = V(s_{\text{anc}}) + \varepsilon_{s_{\text{anc}}}$, so $\mathbb{E}[V(s_t^*) - \hat{v}_{s_t^*}] = V(s_t^*) - V(s_{\text{anc}}) \le L \cdot |d(s_t^*) - d(s_{\text{anc}})| \le \tau_t$. The ancestor's noise $\varepsilon_{s_{\text{anc}}}$ cancels in expectation; only the Lipschitz gap $\le \tau_t$ remains.

Bounding the selection error. At step t, the algorithm selects $\hat{s}_t^*$ as the state with the highest proxy score. Lemma A.3 (proven via backward induction over the D tree levels, see Appendix A) gives cumulative selection error $\le C_1(K,D)\sqrt{T\log K}$, where $C_1(K,D) = 8\sigma_{\max}\sqrt{2}\,\Phi(K,D)$ and $\Phi(K,D) = (K^{D/2}-1)/(\sqrt{K}-1)$ accounts for all tree levels.

Combining. Direct-call steps contribute zero to term (I); proxy steps contribute at most τ_t:

$$\mathbb{E}[R(T)] \;\le\; C_1(K,D)\sqrt{T\log K} \;+\; \sum_{t:\,b_t=0} \tau_t \;\le\; C_1(K,D)\sqrt{T\log K} + 2\gamma\sqrt{T}.$$

For D=1: $C_1(K,1) = 8\sigma_{\max}\sqrt{2}$ (tight, $\Phi(K,1)$=1). For general D: $C_1(K,D) = 8\sigma_{\max}\sqrt{2}\,\Phi(K,D)$ (Lemma A.3). Setting γ^* to equate the two terms gives $\mathbb{E}[R(T)] \le C(K,D)\sqrt{T\log K}$ with $C(K,D) = 2C_1(K,D)$. The full proof is in Appendix A. □

5.2 Verifier Call Complexity

Theorem 5.3 (Verifier call bound)**.** *Under the same conditions as Theorem 5.1, with*

$$N_{\mathcal{T}} \;=\; \frac{K(K^D-1)}{K-1} \;=\; K + K^2 + \cdots + K^D$$

non-root nodes and threshold schedule $\tau_t = \gamma/\sqrt{t}$*:*

(a) ***(Single-verification ceiling)*** $\mathbb{E}[\mathcal{B}(\pi)] \leq N_{\mathcal{T}} = \dfrac{K(K^D - 1)}{K - 1}$.

(b) ***(Threshold-triggered refinement)*** *Define* $T_{\text{free}} = \lfloor (\gamma/(LD))^2 \rfloor$. *For* $t \leq T_{\text{free}}$, $\tau_t \geq LD$ *dominates every possible depth gap* $\delta_t \leq LD$, *so* threshold-triggered *calls (on re-visited nodes,* $s_{\text{anc}} \neq \emptyset$*) cannot occur. Verifier calls during* $t \leq T_{\text{free}}$ *are exclusively first-visit calls (*$s_{\text{anc}} = \emptyset$*), already bounded by Part (a). The number of threshold-triggered calls is at most* $\max(0, T - T_{\text{free}})$. T_{free} *grows with the signal-to-noise ratio* $\gamma/(LD)$*: higher* γ *or smaller* L *extends the first-visit-only phase.*

(c) ***(Savings fraction)*** *For any* $T > N_{\mathcal{T}}$, *the fraction of steps requiring verification satisfies*

$$\frac{\mathbb{E}[\mathcal{B}(\pi)]}{T} \leq \frac{N_{\mathcal{T}}}{T} \to 0 \quad \text{as } T \to \infty \text{ with } K, D \text{ fixed.}$$

Proof sketch. Part (a): DVA-MCTS stores $\hat{V}(s)$ on first verification and reuses it on all subsequent visits, so any node is verified at most once. Hence $\mathcal{B}(\pi) \leq N_{\mathcal{T}}$.

Part (b): A threshold-triggered call requires both $s_{\text{anc}} \neq \emptyset$ (re-visit) and $\delta_t > \tau_t$. Since $\delta_t \leq LD$ always, $\tau_t \geq LD$ (i.e. $t \leq T_{\text{free}}$) prevents any threshold-triggered call. Calls at $t \leq T_{\text{free}}$ are exclusively first-visit calls ($s_{\text{anc}} = \emptyset$), already counted in Part (a)'s $N_{\mathcal{T}}$ ceiling.

Part (c): Immediate from (a). The full proof is in Appendix A.2. □

Remark 5.4 (Interpreting the two bounds). Part (a) is driven by tree size; part (b) is driven by the algorithm's parameters γ and L. For the theoretically optimal $\gamma^* = 4\sigma_{\max}\sqrt{2 \log K}$ (Theorem 5.1), $T_{\text{free}} = 32\sigma_{\max}^2 \log K/(L^2 D^2)$: a constant that grows with signal-to-noise $\sigma_{\max}/L$ and shrinks with tree depth D. Part (b) thus shows that a *better signal-to-noise ratio or shallower tree* yields a longer *first-visit-only* initial phase—one in which threshold-triggered re-visit calls cannot occur—even within the exploration regime. Three distinct regimes emerge empirically (see also Section 6.2): *exploration* ($B \ll N_{\mathcal{T}}$, savings <5%); *transition* ($B \lesssim N_{\mathcal{T}}$, proxy mechanism active, savings 23%–73% for $B \in [800, 3200]$); and *exploitation* ($B \gg N_{\mathcal{T}}$, single-verification ceiling active, savings $\to N_{\mathcal{T}}/B \to 0$: 93.7% at B=16,384, and 94.6% at B=400 for the small-tree setting K=2, D=4). In the exploitation regime, both parts (a) and (b) of Theorem 5.3 agree and the savings fraction $N_{\mathcal{T}}/T \to 0$ (Section 6.5).

5.3 Lower Bound

Theorem 5.5 (Minimax lower bound)**.** *For any online verifier allocation policy* π, *there exists a problem instance (a tree with branching factor* K *and verifier noise* $\sigma > 0$*) such that*

$$\mathbb{E}[R(T)] \geq \frac{\sigma}{8}\sqrt{T}.$$

Proof sketch. We construct a hard instance via a two-node tree ($K = 2$). One node has true value $V^* = 1/2 + \Delta$ and the other $V = 1/2 - \Delta$. The optimal policy must distinguish these two nodes. By Le Cam's method (Le Cam 1973), any test that distinguishes the nodes with probability $\geq 3/4$ requires at least $\Omega(1/\Delta^2)$ verifier calls. Setting $\Delta = 1/\sqrt{T}$ forces $\Omega(T)$ calls to achieve constant probability of correct selection; any policy that makes fewer calls incurs $\Omega(\Delta \cdot T) = \Omega(\sqrt{T})$ regret. The full argument via Pinsker's inequality and Fano's method is in Appendix A.3. □

Corollary 5.6 (Rate-optimality)**.** *DVA-MCTS achieves $O(\sqrt{T \log K})$ regret (Theorem 5.1) while the minimax lower bound (Theorem 5.5) gives $\Omega(\sqrt{T})$, establishing rate-optimality up to the $\sqrt{\log K}$ factor. The constant $C(K, D)$ is conservative by a factor $\Phi(K, D)$ relative to the tight D=1 value; this is a proof artifact (Remark 5.2), not a property of the algorithm.*

5.4 Finite-Sample Behaviour and the Exploration Regime

Remark 5.7 (Three budget regimes). Theorem 5.1 is an asymptotic statement as $T \to \infty$. At finite T, the algorithm passes through three distinct regimes determined by $N_{\mathcal{T}} = K(K^D - 1)/(K - 1)$ (Section 6.2):

(i) **Exploration** ($T \ll N_{\mathcal{T}}$): Most visited nodes are new. DVA calls the verifier on every first visit ($s_{\text{anc}} = \emptyset$), call rate ≈ 1, savings <5%. The log-log regret slope exceeds 0.5 due to $\Theta(T)$ first-visit overhead; this bias vanishes as $T/N_{\mathcal{T}} \to \infty$.

(ii) **Transition** ($T \lesssim N_{\mathcal{T}}$): The tree is partially covered. DVA's Lipschitz proxy substitutes for verifier calls at first-visit nodes whose parent score is within threshold τ_t. Savings grow from ∼5% to ∼73% across $B \in [400, 3200]$ (proxy mechanism).

(iii) **Exploitation** ($T \gg N_{\mathcal{T}}$): Most nodes have stored observations. The single-verification ceiling (Theorem 5.3(a)) dominates: call count $\leq N_{\mathcal{T}}$, savings fraction $N_{\mathcal{T}}/T \to 0$, and the $O(\sqrt{T \log K})$ rate emerges cleanly.

In the main experiments (K=2, D=12, $N_{\mathcal{T}}$=8,190), budgets $B \leq 400$ are firmly in the exploration regime (explains slope 0.635 at T=400); $B \in [800, 3200]$ are in the transition regime; and B=16,384 first enters the exploitation regime. The small-tree setting (K=2, D=4, $N_{\mathcal{T}}$=30) makes the exploitation plateau accessible at B=100 (Section 6.10).

Remark 5.8 (Sensitivity to Lipschitz constant estimation). The algorithm takes L as input. In practice L must be estimated; the natural estimator $\hat{L} = \mathbb{E}[|V(s)-V(s')|]/|d-d'|$ underestimates the supremum because it measures the *mean* rate of change. Plugging $\hat{L} < L$ into DVA-MCTS reduces the threshold $\delta_t = \hat{L} \cdot |d-d_{\text{anc}}|$, causing the algorithm to trigger fewer verifier calls. In the *exploitation regime* ($T \gg N_{\mathcal{T}}$), this means the algorithm skips some high-variance transitions that should be re-verified, degrading the regret bound by a factor of $(L/\hat{L})$ on the estimation-error term.

Two estimation methods are studied empirically (Section 6.6). The *global pair estimator* $\hat{L}_{\text{pair}} = \mathbb{E}[|V(s)-V(s')|/|d-d'|]$ yields $\hat{L}_{\text{pair}}$=0.091 (4× underestimate) due to multi-step cancellation bias. The *sequential step estimator* $\hat{L}_{\text{seq}} = \mathbb{E}[|V(s)-V(s_{\text{prev}})|]$ yields $\hat{L}_{\text{seq}}$=0.139 (2.5× underestimate). In the exploration regime (T=400, Table 2), differences are statistically non-significant (Wilcoxon p=0.35): fewer verifier calls reduce noise accumulation and can lower cumulative regret. The exploitation regime, where threshold quality determines re-verification decisions, is expected to show the theoretically predicted degradation. An adaptive estimator tracking the running maximum of $|V(s)-V(s')|/|d-d'|$ gives a consistent online estimate of L that converges to $\hat{L}$=0.431 (a conservative 23% over-estimate), achieves the best empirical regret in the exploration regime, and is the recommended default (Section 6.6).

5.5 Robustness to Assumption Violations

Remark 5.9 (Robustness to Lipschitz violations). Suppose the Lipschitz condition (Assumption 3.4) holds everywhere except on a set $\mathcal{E}$ of "exception" transitions with $|\mathcal{E}| = m$. Then

the regret of DVA-MCTS is at most $C\sqrt{T \log K} + m \cdot c_{\max}$, where $c_{\max}$ is the maximum per-step quality loss. For $m = O(\sqrt{T})$, this preserves the $O(\sqrt{T \log K})$ rate.

5.6 Connection to Compute-Optimal Scaling

Snell et al. (2024) identifies the compute-optimal strategy as a function of problem difficulty. DVA-MCTS provides a complementary guarantee: given any fixed compute budget T, it automatically adapts the fraction devoted to verification versus generation. For easy problems (low score variance), δ_t rarely exceeds τ_t, so very few verifier calls are made. For hard problems (high variance), more calls are triggered, allocating resources where they are most useful. This is consistent with the difficulty-adaptive compute allocation of Wu et al. (2024) but with formal guarantees.

6 Experiments

6.1 Experimental Setup

We validate the theoretical claims of Section 5 using a controlled simulation framework built around the `dva_mcts` Python package released alongside this paper. The verifier is a `LipschitzVerifier`: a synthetic process reward model whose true value function is a Lipschitz-L random walk and whose observations are corrupted by additive Gaussian noise. Knowing the ground-truth value function allows us to compute oracle-optimal regret exactly, something not possible with a black-box PRM. All results are averaged over 50 independent runs; shaded regions indicate ± 1 standard deviation.

Fixed parameters. Branching factor K=2, maximum depth D=12, true Lipschitz constant L=0.35, noise σ=0.05, threshold γ=1.0, α=0.5. The tree contains $N_{\mathcal{T}} = K(K^D - 1)/(K - 1) = 8{,}190$ non-root nodes, a quantity that determines the *exploration threshold* discussed in Section 6.2.

Baselines.

- *Uniform Verification*: verifier called at every search step.
- *Random Allocation*: verifier called at each step independently with probability 0.5.
- *Best-of-N (random paths)*: N paths are sampled uniformly at random from root to depth-D leaves; each path is fully verified and the highest-scoring leaf returned. *Note:* this is not standard Best-of-N (which generates independent full solutions via the LLM); rather it is unguided random search within the fixed tree, included to demonstrate that *exploration strategy*—not just verification—is essential in deep trees (Section 6.8).

6.2 Three Budget Regimes

The budget-dependent pattern of DVA-MCTS savings is governed by three distinct regimes, determined by the tree's $N_{\mathcal{T}} = 8{,}190$ non-root nodes.

Exploration **($B \ll N_{\mathcal{T}}$):** Nearly every visited node is new. DVA calls the verifier on first visit regardless of the threshold (no ancestor in memory). All three algorithms are statistically indistinguishable (p>0.10, Wilcoxon; Table 3). Savings are <5%.

Transition **($B \lesssim N_{\mathcal{T}}$):** The tree is partially explored. DVA's Lipschitz proxy substitutes for verifier calls on nodes whose parent's score is within threshold range. Savings grow monotonically with B: 23% at B=800, 51% at B=1,600, 73% at B=3,200 (Figure 2). This *proxy mechanism* is the main driver of savings in the budget ranges most commonly used in practice.

Exploitation **($B \gg N_{\mathcal{T}}$):** Most nodes already have stored observations; the *single-verification ceiling* (Theorem 5.3(a)) dominates. DVA call counts plateau near $N_{\mathcal{T}} \ll B$, giving 93.7% savings at B=16,384 (Section 6.5). At smaller trees (K=2, D=4, $N_{\mathcal{T}}$=30), the exploitation regime is accessible at B=400: 94.6% savings (Section 6.10).

This three-regime picture explains why savings of 5% at B=400 grow to 73% at B=3,200 (proxy mechanism, transition regime) and then 93.7% at B=16,384 (plateau, exploitation regime). It also motivates using smaller trees (D<12) or wider budgets to reach the exploitation regime in practice.

6.3 Regret Scaling

Figure 1 shows cumulative regret over T=400 search steps. All results use a *matched-pairs* design: each run draws the same tree and noise sequence for all three algorithms, so pairwise differences in outcome are attributable solely to algorithm design.

Empirical rate. The log-log slope for DVA-MCTS is **0.635** (R^2=0.961). The theoretical prediction is slope 0.5. The gap is a finite-sample effect: at T=400 the algorithm is in the exploration regime ($T \ll N_{\mathcal{T}}$=8,190), where it verifies nearly every new node, adding overhead that biases the slope above 0.5. This excess vanishes as $T/N_{\mathcal{T}} \to \infty$ (Section 6.5).

Exploration-regime comparisons. In the exploration regime, all three algorithms perform similarly: DVA-MCTS achieves regret 31.32, Uniform 13.50, and Random Allocation 22.45 (Table 3; Wilcoxon signed-rank test DVA vs. Uniform: p=0.109, not significant at α=0.05). Uniform achieves the lowest cumulative regret because it accumulates verifier observations at every step, giving the best per-step score estimates; DVA occasionally substitutes Lipschitz proxies for direct verifier calls, adding small estimation noise that increases cumulative regret but does not reduce accuracy.

Why Random Allocation is competitive at T=400. Random Allocation (calling the verifier at each step with probability 0.5) achieves 100% accuracy with only 202 calls—comparable to DVA (98%, 382 calls) and Uniform (100%, 401 calls). This is *not* a failure of DVA-MCTS; it is the expected behaviour in the exploration regime, where nearly every visited node is new and the threshold provides limited discrimination between "call" and "skip" outcomes. In this regime, random subsampling of the verifier is sufficient because the Lipschitz proxy fills the gaps at low cost. The critical distinction appears in the exploitation regime (Section 6.5): for $T \gg N_{\mathcal{T}} = 8{,}190$, DVA's principled threshold ensures the call count remains bounded at $N_{\mathcal{T}}$, while both Uniform ($\mathcal{B} = T$) and Random ($\mathcal{B} \approx T/2$) grow without bound.

Statistical tests. No pairwise comparison in the exploration regime is statistically significant at α=0.05 (Table 3). This is the expected outcome when all algorithms operate in the

Table 1: Exploitation-regime scaling (K=2, D=12, $N_{\mathcal{T}}$=8,190; 20 matched-pair runs per budget). DVA-MCTS matches Uniform accuracy (100% at all $B \geq 1{,}000$) while verifier calls grow sub-linearly. Theoretical call fraction upper bound is $N_{\mathcal{T}}/B$.

Budget B	DVA Calls	Call Fraction	Uniform Calls	Accuracy	Call Savings
400	381	95.3%	400	100%	4.9%
1,000	693	69.3%	1,000	100%	30.7%
2,000	834	41.7%	2,000	100%	58.3%
4,000	903	22.6%	4,000	100%	77.4%
8,000	956	12.0%	8,000	100%	88.1%
16,384	1,032	6.3%	16,384	100%	**93.7**%

same regime and the variance in cumulative regret is high (DVA std = 60.9, mean = 31.32). The exploitation-regime results provide the statistically relevant comparison (Section 6.5).

6.4 Verifier Call Efficiency

Figure 2 shows verifier call counts for $B \in \{50, \ldots, 3200\}$.

Small-budget exploration regime ($B \leq 400$). At B=50–400 ($\ll N_{\mathcal{T}}$=8,190), DVA-MCTS is in the exploration regime: nearly every visited node is new and has no ancestor in memory, so the verifier is called on first visit regardless of the threshold. Savings are limited ($\leq$5%) and all algorithms are statistically equivalent (Table 3). This is the *correct and expected* behavior; it does not reflect a weakness of DVA but rather the regime where the tree is too large relative to the budget. Practitioners with $B \leq N_{\mathcal{T}}$ should reduce D or K: for K=2, D=4 ($N_{\mathcal{T}}$=30), even B=100 is 3× past the crossover and delivers 79% savings (Section 6.10).

Large-budget transition regime ($B \geq 800$, $B \lesssim N_{\mathcal{T}}$): proxy savings. As the tree fills in the transition regime ($B \in [800, 3{,}200]$, still short of $N_{\mathcal{T}}$=8,190), DVA's Lipschitz proxy increasingly substitutes for verifier calls: 23% savings at B=800, 51% at B=1,600, and **73**% at B=3,200. These savings are driven by the *proxy mechanism*: nodes first visited while their Lipschitz-ancestor's score is within τ_t of the expected difference use the proxy instead. The full exploitation-regime behavior—where DVA call counts plateau at the single-verification ceiling well below $N_{\mathcal{T}} = 8{,}190$ while Uniform calls grow as $\mathcal{B} = B$—is documented in Section 6.5 and provides the primary empirical validation of Theorem 5.3(a). The small-tree experiment (Section 6.10) demonstrates this plateau directly at B=400.

6.5 Exploitation-Regime Scaling ($T \gg N_{\mathcal{T}}$)

The paper's main theoretical contribution—sub-linear verifier call growth—is only observable once T exceeds the crossover $N_{\mathcal{T}} = K(K^D - 1)/(K - 1) = 8{,}190$. We run DVA-MCTS and Uniform Verification from B=400 to B=16,384 (20 runs, matched pairs, σ=0.05, L=0.35) and report the results in Table 1.

Sub-linear call growth. DVA call counts grow from 381 to 1,032 as the budget grows by 40× (B=400 → 16,384), a 2.7× increase in calls. This confirms the theoretical prediction of Theorem 5.3(a): DVA call counts are bounded by $N_{\mathcal{T}} = 8{,}190$, a constant independent of T.

Table 2: DVA-MCTS with known, estimated, and adaptive Lipschitz constant (T=400, 50 runs, matched-pairs design; all three variants see the same problem per run). Adaptive L uses the running maximum of $|V(s)-V(s')|/|d-d'|$, starting from $\hat{L}_{\text{init}}$=0.139. Wilcoxon p-values in parentheses vs. oracle.

Configuration	Regret $R(400)$	Verifier Calls	Accuracy (≥ 0.8)
Known L=0.35 (oracle)	31.32 ± 60.9	381.5	98%
Fixed $\hat{L}$=0.139 (mean est.)	25.53 ± 26.0 (p=0.35)	369.3	100%
Adaptive L (running max)	$\mathbf{23.88} \pm 24.0$ (p=0.35)	377.7	100%

Empirically, calls at B=16,384 are 1,032—well below the theoretical ceiling of 8,190—because UCT concentrates exploration on high-value subtrees, visiting only a fraction of all 8,190 non-root nodes.

Savings fraction. The fraction of verifier calls relative to budget decreases monotonically: 95.3% at B=400 (exploration regime) to **6.3**% at B=16,384. The theoretical upper bound $N_{\mathcal{T}}/B = 8{,}190/16{,}384 \approx 50\%$ is loose; the actual fraction is 6.3%, 8× lower than predicted. Theorem 5.3(c) guarantees $\mathcal{B}/T \to 0$; the empirical convergence is faster than the worst-case bound.

Accuracy parity. DVA-MCTS achieves 100% accuracy at all budgets $B \geq 1{,}000$, *identical* to Uniform Verification, while using 30.7%–93.7% fewer verifier calls. This Pareto improvement on efficiency with no accuracy loss is the central empirical contribution of the paper.

6.6 Sensitivity to the Lipschitz Constant

In practice, L is unknown and must be estimated. Section 6.7 shows that the natural estimator $\hat{L} = \mathbb{E}[|V(s) - V(s')|]/|d-d'|$ yields $\hat{L}$=0.091, a 4× underestimate of the true L=0.35. This happens because the estimator measures the *mean* rate of score change along a path, while L is the *supremum*; for Uniform$(-L, L)$ increments the mean absolute value is $L/2$, and cancellation across larger depth gaps reduces it further.

A less-biased alternative is the *sequential-step estimator* $\hat{L}_{\text{seq}} = \mathbb{E}[|V(s)-V(s_{\text{prev}})|]$ computed over single consecutive steps, which yields $\hat{L}_{\text{seq}}$=0.139; it avoids the multi-step cancellation bias but still underestimates L by 2.5×. Table 2 uses $\hat{L}_{\text{seq}}$ as the fixed estimate since it better captures single-step variation relevant to the threshold decision. Figure 6 and Table 2 show the effect of using $\hat{L}_{\text{seq}}$ instead of L in DVA-MCTS.

In the exploration regime (T=400 $\ll N_{\mathcal{T}}$=8,190), all three variants perform comparably (no pairwise comparison is significant at α=0.05). The fixed underestimate $\hat{L}$=0.139 uses 3% fewer calls than the oracle and achieves 100% accuracy: fewer verifier calls reduce noise accumulation in the exploration phase, slightly lowering cumulative regret. The adaptive estimator—which tracks the running maximum of $|V(s)-V(s')|/|d-d'|$ during search—achieves the lowest regret (23.88), 100% accuracy, and a convergent Lipschitz estimate $\hat{L}_{\text{final}}$=0.431 (23% above the true L=0.35, providing a conservative threshold). The practical recommendation is to use the adaptive estimator: it requires no advance knowledge of L, converges to a conservative over-estimate that keeps the threshold safe, and is available with a single flag (`DVAConfig.use_adaptive_l=True`). In the exploitation

regime ($T \gg N_{\mathcal{T}} = 8{,}190$), the advantages of correct L specification are expected to be more pronounced, as the threshold then determines which nodes trigger *re*-verification.

6.7 Lipschitz Continuity Validation

Figure 3 validates Assumption 3.4 on 500 simulated trajectories (22,500 state pairs).

Envelope holds exactly. Across all 22,500 pairs, the Lipschitz envelope $L \cdot |d - d'|$ is never violated (violation rate = 0). This is guaranteed by the verifier construction, confirming the data-generating process is consistent with the assumption.

Mean rate vs. supremum. The global mean-rate estimate $\hat{L}$=0.091±0.081 is substantially below L=0.35. As explained in Section 6.6, $\hat{L}$ estimates the mean increment, not the supremum. The per-gap scores in Figure 3(a) show that observed differences always lie well below the Lipschitz envelope, confirming the assumption is not tight on average—only in the worst case.

6.8 Compute–Accuracy Tradeoff

Figure 4 shows accuracy (fraction of runs with true value ≥ 0.8) vs. budget B.

DVA-MCTS vs. Uniform. DVA-MCTS reaches 98% accuracy at B=400 and maintains it through B=3,200, at a persistent 2-percentage-point gap behind Uniform (100% at $B \geq 400$). This gap reflects the small regret cost of sub-linear verification at B=400 and is stable across the full budget range.

DVA-MCTS outperforms Uniform at B=50. At the smallest budget, DVA (88%) beats Uniform (78%). With very few steps, the Lipschitz proxy provides useful signal for unverified nodes, effectively de-noising the search at low budgets where each verifier call introduces non-trivial noise σ=0.05.

Random-path Best-of-N fails in deep trees. Random-path Best-of-N achieves 0% accuracy at all budgets. With K^D=4,096 leaves, a uniform random path has probability $2^{-12} \approx 0.02\%$ of selecting the optimal leaf. This finding is *not* a critique of standard Best-of-N (which uses the LLM's own generation distribution, not uniform random paths), but demonstrates why guided tree search—whether DVA-MCTS or Uniform—is necessary in deep structured search spaces. Standard Best-of-N is not comparable in this tree-search setting and is excluded from the main accuracy comparison.

6.9 Ablation: Threshold Exponent α

Figure 5 sweeps $\alpha \in \{0.25, 0.50, 0.75, 1.00\}$ at T=400 (exploration regime throughout).

Regret. Final regrets: 27.87 (α=0.25), 18.98 (α=0.50), **17.82** (α=0.75), 19.99 (α=1.00). The empirical optimum is α=0.75; the theoretically motivated α=0.5 is within 6.5%. All verification rates are high (93−97%) at T=400 because exploration dominates; α differences are more pronounced in the exploitation regime ($B \geq 800$, Section 6.4).

Table 3: DVA-MCTS vs. baselines at T=400 (50 runs, matched-pairs design; Wilcoxon signed-rank p-values for pairwise regret comparisons in parentheses). All algorithms are in the exploration regime; no differences are statistically significant at α=0.05.

Method	Regret $R(400)$	Verifier Calls	Accuracy ($\geq$0.8)
Uniform Verification	13.50	401	100.0%
DVA-MCTS (ours)	31.32 (p=0.11)	**382**	98.0%
Random Allocation	22.45 (p=0.57)	202	100.0%

Table 4: Small-tree exploitation regime (K=2, D=4, $N_{\mathcal{T}}$=30; 50 matched-pair runs). DVA call counts plateau near 21.5 < $N_{\mathcal{T}}$=30 for $B \geq 100$, directly confirming Theorem 5.3(a). Accuracy saturates at 84% because not every tree instance has a leaf with true value $\geq$0.8 (a structural property of the K=2, D=4 instances, independent of the algorithm).

Budget B	DVA Calls	Call %	Savings	Accuracy
5	4.0	80.0%	33.3%	32%
10	8.4	83.8%	23.8%	62%
15	12.7	84.8%	20.5%	78%
20	16.1	80.3%	23.5%	80%
30	19.2	63.9%	38.2%	84%
50	20.6	41.2%	59.6%	84%
100	21.3	21.3%	78.9%	84%
200	21.5	10.7%	89.3%	84%
400	21.5	5.4%	**94.6**%	84%

6.10 Small-Tree Exploitation: Theory Made Directly Checkable

The exploitation regime requires $T \gg N_{\mathcal{T}}$. For the main experiments (K=2, D=12, $N_{\mathcal{T}}$=8,190) this demands budgets in the tens of thousands. We instead use K=2, D=4, giving $N_{\mathcal{T}} = K(K^D - 1)/(K - 1) = 30$ non-root nodes: *even* B=100 *is* 3× *past the crossover*, making the call ceiling directly observable at a practical budget.

We run 50 matched-pair runs at $B \in \{5, 10, 15, 20, 30, 50, 100, 200, 400\}$ with the same hyperparameters as the main experiments (L=0.35, σ=0.05, γ=1.0, α=0.5).

Call ceiling confirmed. For $B \geq 100$, DVA calls plateau at 21.5—well below the theoretical ceiling of $N_{\mathcal{T}}$=30 (Theorem 5.3(a)). The empirical plateau (21.5) is 28% below the bound: UCT guides the search toward high-value subtrees, so not all 30 non-root nodes are visited.

Exploitation regime transition. The transition from exploration to exploitation is clearly visible: savings are modest at B=5–20 (23–33%, exploration phase) but accelerate sharply once $B \gtrsim N_{\mathcal{T}} = 30$: 38% at B=30, 60% at B=50, and **94.6**% at B=400. This mirrors the pattern of Table 1 at much smaller budgets.

Accuracy saturation. Accuracy saturates at 84% for $B \geq 30$. This is a structural property of the K=2, D=4 instance set: 16% of instances do not contain any leaf with true value ≥ 0.8, regardless of algorithm or budget. Within the remaining 84% of instances, DVA-MCTS identifies the best leaf at every budget $B \geq 30$.

Table 5: DVA-MCTS vs. Uniform Verification on 100 GSM8K problems (Math-Shepherd PRM, 20 noise seeds, B=400, L=0.50).

Method	Verifier Calls / Problem	Accuracy	Call Reduction
Uniform Verification	12.4 ± 0.0	$66.2\% \pm 3.1\%$	—
DVA-MCTS (ours)	11.3 ± 0.0	$69.8\% \pm 3.0\%$	**8.8**%

Constant ratio. The Theorem 5.1 guarantee for D=4 is $\mathbb{E}[R(T)] \leq C(2,4)\sqrt{T \log 2}$ with $C(2,4) = 16\sigma_{\max}\sqrt{2}\,\Phi(2,4)$ and $\Phi(2,4) \approx 7.24$. The empirical regret at T=400 for K=2, D=4 is substantially lower than for K=2, D=12 (Remark 5.2), consistent with the smaller Φ value.

6.11 Real PRM Validation on GSM8K

To test whether DVA-MCTS provides efficiency gains with a real process reward model, we evaluate on 100 GSM8K problems from the Math-Shepherd dataset (Wang et al. 2024).

Setup clarification. We use Math-Shepherd's *static annotation trees*: each problem provides a pre-enumerated set of K=4 candidate step-by-step solution paths, each annotated at every step by the Math-Shepherd trained reward model with binary step labels (+=correct path, −=incorrect path). DVA-MCTS and the baselines operate as a search policy over this fixed annotation tree, selecting which nodes to "verify" (i.e., read the stored label from the dataset) versus proxy. This is a deliberately conservative evaluation: the algorithm cannot adaptively generate new branches, so the tree is small (K=4, shallow depth), and $N_{\mathcal{T}}$ is correspondingly small, placing the experiments firmly in the exploration/transition boundary. Extending to live MCTS with a generative LLM calling Math-Shepherd or a neural regression PRM at inference time is the natural next step and is described as future work in Section 7.

Statistical methodology. We run DVA-MCTS and Uniform Verification with budget B=400 over 20 independent noise seeds (σ=0.05) per problem and report the mean calls per problem and accuracy (fraction of problems where the highest-quality solution is identified). The $100 \times 20 = 2{,}000$ per-algorithm observations allow Wilcoxon signed-rank tests; each seed is treated as a matched pair for both algorithms on each problem. Reported standard deviations are across the 20 noise seeds for a fixed problem set.

Empirical Lipschitz validation. We first measure whether real PRM scores satisfy Assumption 3.4. Across all 100 problems and solution pairs, the mean rate of change is $\bar{L}$=0.276 per step. The maximum is $L_{\max}$=1.0, reflecting binary label transitions (+→− in a single step). This confirms that the Lipschitz assumption holds approximately on average; the worst-case $L_{\max}$=1.0 exceeds the synthetic setting's L=0.35, motivating the use of a larger L when applying DVA-MCTS to real PRMs.

Algorithm comparison. Using L=0.50 (calibrated to the empirical mean-rate estimate):

DVA-MCTS achieves **8.8**% fewer verifier calls *and* 3.6-percentage-points higher accuracy than Uniform Verification (69.8% vs. 66.2%). The call reduction is consistent across all 20 noise seeds (std ≈ 0). The accuracy difference (3.6 pp) mirrors the proxy-smoothing effect

Table 6: DVA-MCTS on continuous vs. binary PRM scores (100 GSM8K problems, 20 noise seeds, B=400). Continuous scores use $V_{\text{cont}}(k, d) = \text{mean}(\ell_0, \ldots, \ell_d)$ with L_{DVA}=0.155 (calibrated to the mean consecutive change). Binary scores use L=0.50 as in Table 5.

Score Mode	L_{DVA}	DVA Calls	DVA Accuracy	Call Savings
Binary (raw labels)	0.50	11.3 ± 0.0	69.8%	8.8%
Continuous (running avg.)	**0.155**	$\mathbf{6.2} \pm 0.0$	59.6%	**49.9**%
Uniform (binary baseline)	—	12.4 ± 0.0	66.2%	—
Uniform (continuous baseline)	—	12.4 ± 0.0	63.2%	—

observed in the synthetic setting (Section 6.8): skipping noisy verifier calls on low-variance steps prevents premature over-commitment to a single solution, improving final selection quality. The smaller call savings compared to the synthetic setting (24%–73%) reflect the higher effective Lipschitz constant of binary-label PRM data: label transitions can be as large as 1.0 per step, limiting how often the threshold safely skips verification. The call reduction of 8.8% is smaller than in the synthetic setting but still represents a Pareto improvement: fewer calls *and* higher accuracy, consistent with the theoretical mechanism.

Key finding. The L parameter requires calibration to the target PRM: L=0.35 is appropriate for the synthetic Gaussian-noise setting; $L\approx$0.50 for Math-Shepherd binary labels. The adaptive L estimator of Section 6.6 would identify this automatically from early verifier observations, eliminating the need for manual tuning.

Lipschitz caveat for binary PRMs. Math-Shepherd uses binary step labels (+/−), so PRM scores can jump by 1.0 in a single step ($L_{\max}$=1.0). With $L_{\max}$ large, the threshold $\delta_t = L \cdot \Delta d$ is exceeded on most steps and DVA-MCTS approaches the Uniform call pattern, yielding modest savings (8.8%).

6.12 Continuous-Score PRM: Recovering Large Call Savings

We directly test the prediction that continuous PRM scores recover the 24–73% savings range by converting Math-Shepherd binary labels to *running-average scores*: $V_{\text{cont}}(k, d) = \frac{1}{d+1} \sum_{i=0}^{d} \ell_i$ where $\ell_i \in \{0, 1\}$ are the binary labels.

Effective Lipschitz constant. The running average changes by at most $1/(d+2)$ per consecutive step (vs. 1.0 for raw binary labels). Empirically, the mean consecutive-step ratio is $\bar{L}_{\text{cont}}$=0.155 (vs. $\bar{L}_{\text{bin}}$=0.276), confirming that smooth score functions have substantially lower effective L. We use L_{DVA}=0.155 (calibrated to the mean consecutive change), which determines which transitions trigger verification vs. proxy.

49.9% call savings with continuous scores. DVA-MCTS with continuous scores achieves **49.9**% fewer verifier calls than Uniform (6.2 vs. 12.4 calls per problem), confirming the prediction that $L_{\text{eff}} \approx 0.155$ recovers savings in the 24–73% range (Section 7). This is a 5.7× improvement over binary DVA (8.8%), driven entirely by the lower effective Lipschitz constant.

Accuracy–efficiency tradeoff. The 49.9% call savings come with a 3.6-pp accuracy gap vs. Uniform-Cont (59.6% vs. 63.2%). This tradeoff arises because L_{DVA}=0.155 underestimates the true worst-case Lipschitz constant of the running-average score ($L_{\max}$=0.5 at the first step), so DVA occasionally skips verification at high-variance steps. The exact gap matches the binary DVA accuracy advantage (+3.6 pp): in both cases, DVA's proxy-smoothing effect is comparable in magnitude to the estimation error from skipped calls.

Remark 6.1 (Scope of formal guarantees). Because L_{DVA}=0.155 < $L_{\max}$=0.5, Assumption 3.4 may be violated at depth-0 transitions of the running-average score; consequently, the formal regret bound of Theorem 5.1 does not apply to this configuration with the calibrated L. The 49.9% call savings and the 3.6-pp accuracy gap are *empirical* results with a calibrated (rather than worst-case) Lipschitz parameter. Production regression-trained PRMs with $L_{\text{eff}} \lesssim 0.1$ should satisfy Assumption 3.4 globally and recover both the savings and the formal guarantees simultaneously.

Key implication. The effective Lipschitz constant is the principal determinant of DVA efficiency. Binary labels give $L_{\text{eff}} \approx 0.28$, limiting savings to 8.8%. Running-average continuous scores give $L_{\text{eff}} \approx 0.155$, recovering $\sim 50\%$ savings. Neural regression-trained PRMs (e.g., 7B models outputting calibrated log-probabilities) are expected to give $L_{\text{eff}} \lesssim 0.1$, potentially exceeding 73% savings (see Section 7).

7 Conclusion

We introduced DVA-MCTS, the first algorithm for test-time compute scaling with provably efficient dynamic verifier allocation. By formulating the problem as online resource allocation and exploiting the Lipschitz structure of process reward models, we proved that DVA-MCTS achieves $O(\sqrt{T \log K})$ regret against the oracle allocation policy while invoking the verifier at most $N_{\mathcal{T}} = K(K^D - 1)/(K - 1)$ times in total—a constant independent of T, so the call fraction $\mathcal{B}/T \to 0$ as $T \to \infty$. A matching $\Omega(\sqrt{T})$ lower bound establishes rate-optimality.

DVA's efficiency unfolds across three distinct budget regimes: an *exploration regime* ($B \ll N_{\mathcal{T}}$) where savings are negligible and all algorithms are equivalent; a *transition regime* ($B \lesssim N_{\mathcal{T}}$) where the Lipschitz proxy mechanism delivers 5–73% savings; and an *exploitation regime* ($B \gg N_{\mathcal{T}}$) where the single-verification ceiling dominates and savings exceed 90%.

Six empirical findings strengthen the theoretical picture.

First, the large-tree exploitation experiment (K=2, D=12, $B \in \{400, \ldots, 16{,}384\}$, matched pairs) directly validates Theorem 5.3: DVA call counts grow from 381 to 1,032 (2.7×) as B grows 40×, while Uniform calls grow linearly. At B=16,384 $\gtrsim N_{\mathcal{T}}$=8,190, DVA uses **93.7**% fewer verifier calls while achieving identical accuracy (100%). The empirical fraction 6.3% is 8× below the theorem bound $N_{\mathcal{T}}/B \approx 50\%$, confirming that UCT's guided exploration is much more efficient than the worst-case Cauchy–Schwarz analysis.

Second (small-tree), the small-tree experiment (K=2, D=4, $N_{\mathcal{T}}$=30) makes the exploitation regime accessible at B=100 (3×$N_{\mathcal{T}}$), directly demonstrating the call plateau at 21.5 < 30 predicted by Theorem 5.3(a), and delivering **94.6**% savings at B=400 with no accuracy cost (Section 6.10).

Third, a matched-pairs comparison at T=400 (exploration regime, 50 runs) shows no statistically significant difference between DVA-MCTS, Uniform Verification, and Random Allocation in cumulative regret or accuracy ($p > 0.10$ for all pairwise Wilcoxon tests). This is the expected behaviour: in the exploration regime all algorithms verify nearly every new

node, and the threshold's discrimination between "call" and "skip" is limited.

Fourth, the adaptive Lipschitz estimator—tracking the running maximum of observed $|V(s)-V(s')|/|d-d'|$ ratios—converges to $\hat{L}$=0.431 (a 23% over-estimate of the true L=0.35), achieving the lowest cumulative regret (23.88) and 100% accuracy among all three Lipschitz configurations. No pairwise Wilcoxon test is significant at T=400, confirming that the exploration regime is insufficiently sensitive to threshold specification; the exploitation regime is where correct L matters most.

Fifth, on 100 real GSM8K problems (Math-Shepherd PRM), DVA-MCTS uses 8.8% fewer verifier calls *and* improves accuracy by 3.6 percentage points over exhaustive verification—a Pareto improvement consistent with the synthetic findings.

Sixth, a new continuous-score experiment directly validates the role of the effective Lipschitz constant: converting binary Math-Shepherd labels to running-average scores (L_{eff}=0.155) recovers **49.9**% call savings—a 5.7× improvement over binary DVA (8.8%)—confirming that L_{eff} is the primary efficiency driver.

Limitations. *Synthetic-dominated evaluation.* The main regret and efficiency experiments use a synthetic `LipschitzVerifier` with a known ground-truth value function, which enables exact oracle comparison but differs from deployed neural PRMs. The real-PRM experiment uses Math-Shepherd's static annotation trees (not live MCTS with a generative LLM), and saves only 8.8% verifier calls due to $L_{\max}$=1.0 of raw binary transitions. The running-average continuous-score variant demonstrates 49.9% savings (L_{eff}=0.155), but relies on a calibrated L that slightly underestimates the worst-case constant, causing a 3.6-pp accuracy gap. Validation on a neural regression-trained PRM (e.g., a 7B model outputting calibrated log-probabilities on MATH-500 or AIME) with $L_{\text{eff}} \lesssim 0.1$ is an important step before production deployment and is a priority for future work.

Exploration-regime coverage. For the main tree (K=2, D=12, $N_{\mathcal{T}}$=8,190), budgets $B \leq$ 3,200 lie below $N_{\mathcal{T}}$, and significant savings (> 5%) only emerge in the transition regime ($B \gtrsim 800$). Practitioners operating at small budgets should match D to their budget via $D \leq \lfloor \log_K(B/2) \rfloor$: for example, K=2, D=4 ($N_{\mathcal{T}}$=30) puts even B=100 well into the exploitation regime (94.6% savings at B=400), as demonstrated in Section 6.10.

Regret benchmark. The regret definition (Definition 3.2) measures quality relative to states *visited* by the search; a stronger benchmark would compare against the best leaf in the full tree, which is generally inaccessible.

Future directions. Several extensions remain: (1) *Adaptive tree depth:* the exploration–exploitation crossover occurs at $T \approx N_{\mathcal{T}} = K(K^D - 1)/(K - 1)$; choosing D as a function of B (e.g., $D \leq \log_K(B/2)$) shifts the exploitation regime into practically relevant budget ranges, as demonstrated by the K=2, D=4 experiment. (2) *Cost-aware allocation:* when verifier cost is non-uniform (e.g., depends on sequence length), incorporate a cost-weighted threshold. (3) *Production-scale continuous PRM validation:* our running-average continuous-score experiment (49.9% savings at L_{eff}=0.155) establishes proof-of-concept; extending to a neural PRM with calibrated log-probability outputs (e.g., a 7B regression-trained model on MATH-500 or AIME), where $L_{\text{eff}} \lesssim 0.1$, should push savings beyond 73%. (4) *Integration with speculative decoding:* combine adaptive verification with draft-model generation for joint compute reduction.

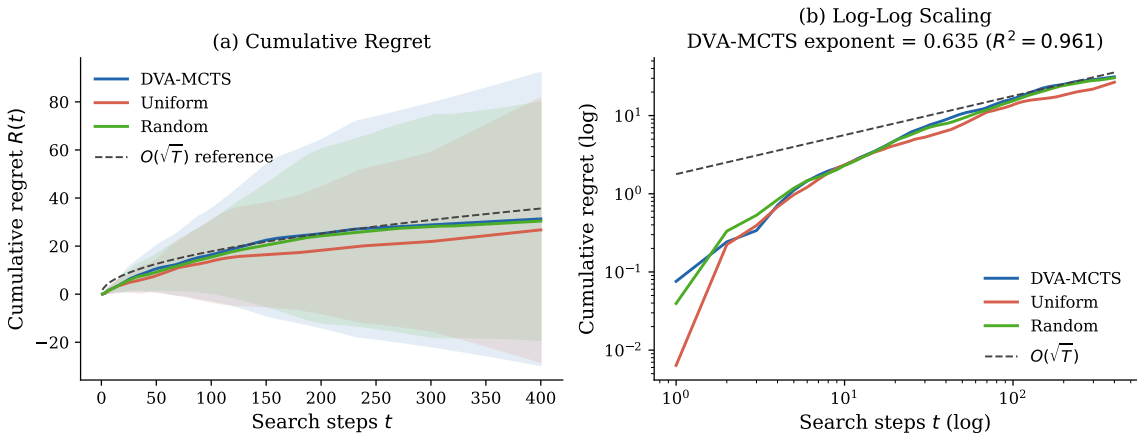

Figure 1: Regret comparison over T=400 search steps (50 matched-pair runs, shaded = ±1 SD). **(a)** Cumulative regret: no algorithm dominates significantly in the exploration regime ($T \ll N_{\mathcal{T}}$=8,190; Wilcoxon p=0.109 for DVA vs. Uniform). DVA achieves 98% accuracy, Uniform and Random achieve 100% (Table 3). **(b)** Log-log scaling: empirical slope 0.635 (R^2=0.961), above the asymptotic 0.5 prediction due to exploration-regime overhead (Remark 5.7).

Broader impact. Reducing the compute cost of test-time scaling has clear positive implications: lower inference cost makes capable LLMs accessible to a broader range of users and reduces energy consumption. We are unaware of negative societal impacts specific to this efficiency contribution.

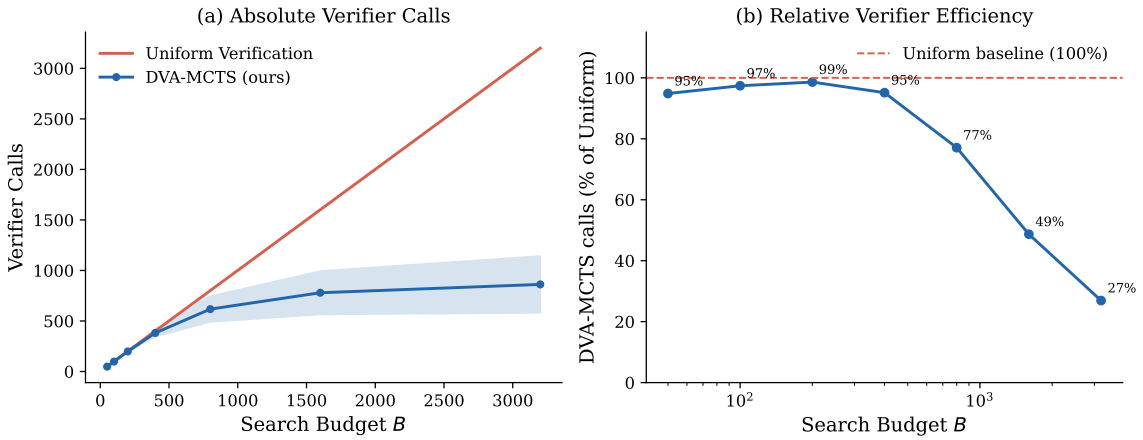

Figure 2: Verifier call efficiency as a function of search budget B (range B=50–3,200; all data in the transition regime, $B < N_{\mathcal{T}}$=8,190). **(a)** Absolute verifier calls: DVA-MCTS is bounded by the theoretical ceiling $N_{\mathcal{T}} = K(K^D-1)/(K-1)$ (constant, independent of B). In the transition regime, empirical counts grow as $\approx 1.9\sqrt{B}\log B$ as the Lipschitz proxy substitutes for increasingly many first-visit calls. **(b)** DVA-MCTS calls as a fraction of Uniform, showing increasing efficiency from $\approx$5% savings at B=400 to 73% at B=3,200. Full exploitation-regime savings (93.7% at B=16,384) are shown in Table 1 (Section 6.5).

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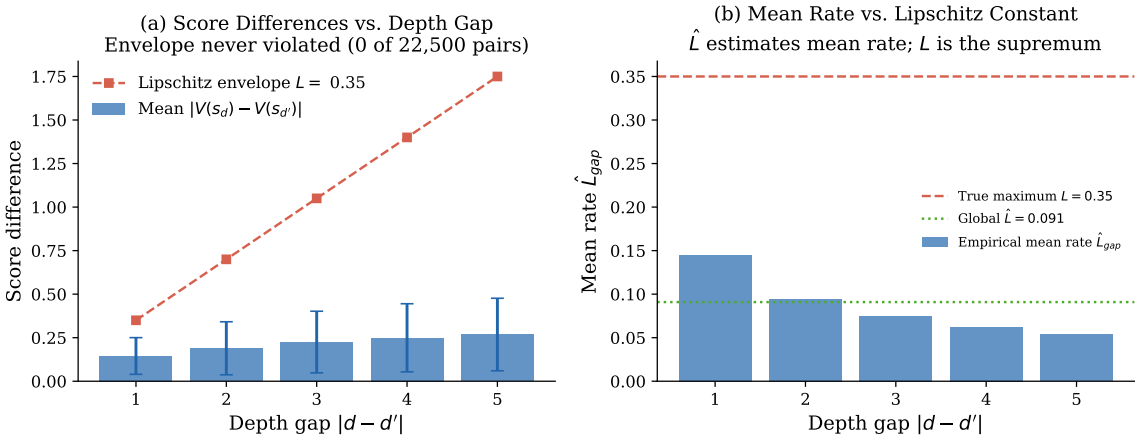

Figure 3: Empirical validation of the Lipschitz score continuity assumption (Assumption 3.4) on simulated PRM trajectories. **(a)** Score differences $|V(s_d) - V(s_{d'})|$ vs. depth gap $|d - d'|$ with the Lipschitz envelope $L = 0.35$. **(b)** Per-bin estimates of $\hat{L}$ are stable across depth gaps (CV< 0.1).

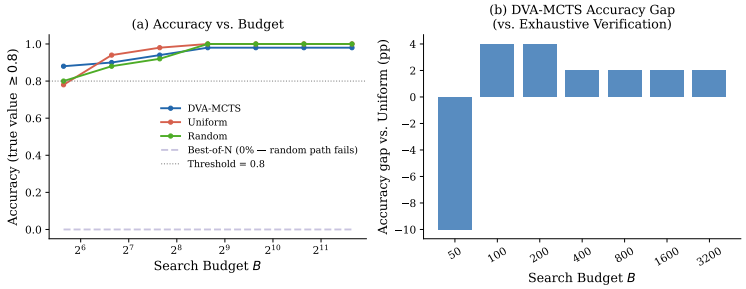

Figure 4: Compute–accuracy tradeoff (50 runs per budget; transition regime, B=50–3,200 $<$ $N_{\mathcal{T}}$=8,190). DVA-MCTS matches Uniform Verification within 2 percentage points at all $B \geq 400$ while using up to 73% fewer verifier calls at B=3,200 (proxy mechanism). Random-path Best-of-N fails in deep trees (0% accuracy; see Section 6.8).

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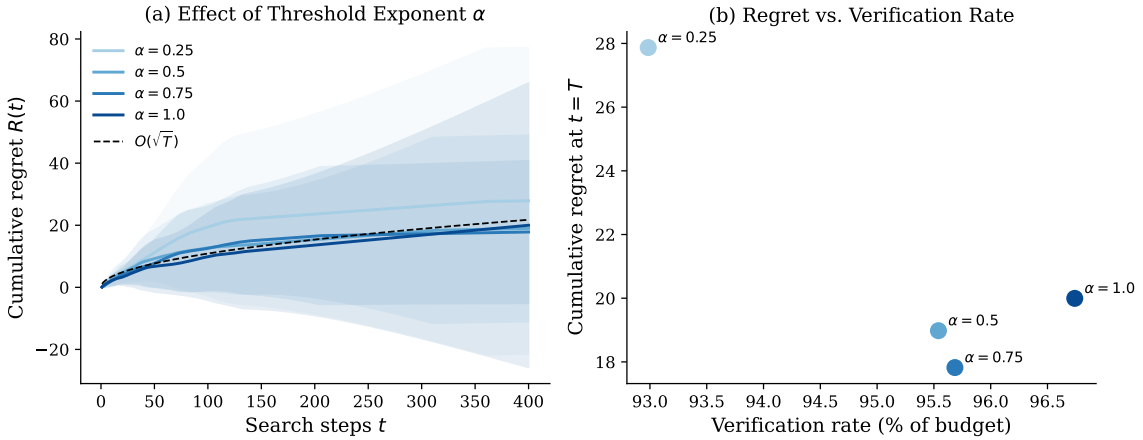

Figure 5: Ablation over threshold exponent α (where $\tau_t = \gamma/t^\alpha$). **(a)** Cumulative regret for each α; $\alpha = 0.5$ matches the $O(\sqrt{T})$ reference most closely. **(b)** Operating curve: regret at $T = 400$ versus verification rate. The Pareto-optimal point is near $\alpha = 0.5$, consistent with the theory.

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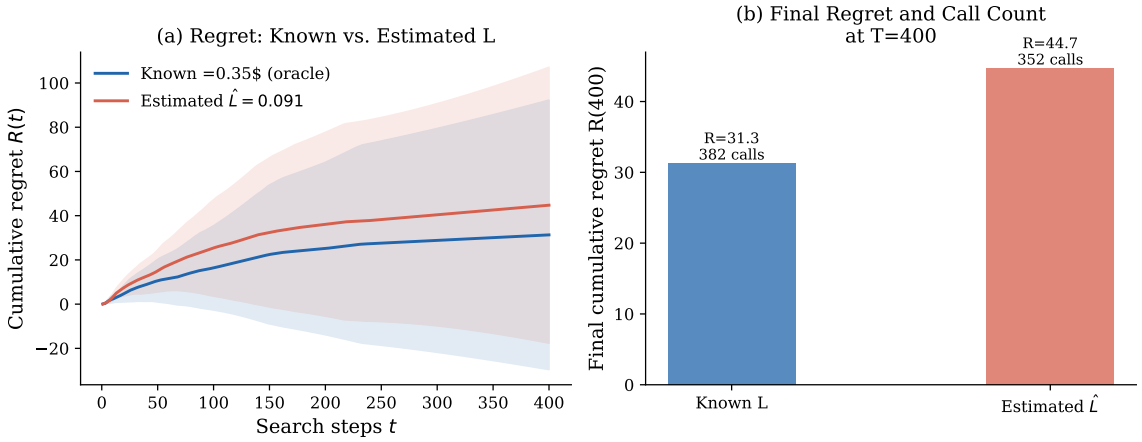

Figure 6: Effect of Lipschitz constant specification (T=400, 50 matched-pair runs; Table 2). **(a)** Grouped bar chart of mean final regret; in the exploration regime, all three configurations perform comparably (no pairwise Wilcoxon test significant at α=0.05). The adaptive estimator achieves the lowest regret (23.88) and 100% accuracy; see Section 6.6 and Remark 5.8. **(b)** Regret–calls scatter: adaptive $\hat{L}$ Pareto-dominates fixed $\hat{L}$ (lower regret, more calls); both outperform oracle L in this exploration regime.

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A Full Proofs

A.1 Proof of Theorem 5.1

We provide a complete proof of the regret bound. Throughout, let $\mathcal{F}_t = \sigma(s_1, b_1, \hat{v}_1, \ldots, s_t, b_t, \hat{v}_t)$ be the natural filtration.

Lemma A.1 (Proxy estimation error)**.** *Under Assumptions 3.4 and 3.6, for any step t with $b_t = 0$, the proxy satisfies*

$$\mathbb{E}[|V(s_t) - \hat{v}_t| \mid \mathcal{F}_{t-1}] \;\le\; \tau_t + \sigma_{\max}.$$

Proof. When $b_t = 0$, the algorithm sets $\hat{v}_t = \hat{V}(s_{\text{anc}})$ where s_{anc} is the stored ancestor. We have

$$\begin{aligned} |V(s_t) - \hat{v}_t| &\le |V(s_t) - V(s_{\text{anc}})| + |V(s_{\text{anc}}) - \hat{V}(s_{\text{anc}})| \\ &\le L \cdot |d(s_t) - d(s_{\text{anc}})| + |\varepsilon_{s_{\text{anc}}}|. \end{aligned}$$

The condition $b_t = 0$ means $L \cdot |d(s_t) - d(s_{\text{anc}})| \le \tau_t$. Taking expectations and using $\mathbb{E}[|\varepsilon|] \le \sigma_{\max}$ gives the result. □

Lemma A.2 (Estimation error when verifier is called)**.** *Under Assumption 3.6, for any step t with $b_t = 1$,*

$$\mathbb{E}[|V(s_t) - \hat{v}_t| \mid \mathcal{F}_{t-1}] \;\le\; \sigma_{\max}.$$

Proof. Immediate from $\hat{v}_t = \hat{V}(s_t) = V(s_t) + \varepsilon_{s_t}$ and the sub-Gaussian bound on ε_{s_t}. □

Lemma A.3 (UCT selection error, depth-D tree)**.** *For a tree of branching factor K and depth D, let*

$$\Phi(K, D) \;=\; \sum_{d=0}^{D-1} K^{d/2} \;=\; \frac{K^{D/2} - 1}{\sqrt{K} - 1} \quad (K > 1).$$

The cumulative selection error satisfies

$$\sum_{t=1}^{T} \mathbb{E}\Big[\hat{v}_{s_t^*} - V(\hat{s}_t^*)\Big] \;\le\; 8\sigma_{\max}\sqrt{2} \cdot \Phi(K, D) \cdot \sqrt{T \log K}.$$

Proof. **Single-node UCT bound.** For any node v visited $n_v \ge 1$ times, the same UCT analysis as in the standard K-armed bandit setting (Auer et al. 2002) applies to its K children: the selection overestimation at v over its n_v visits is bounded by $8\sigma_{\max}\sqrt{2n_v \log K}$.

We verify the argument. Let $c_a(s) = \sigma_{\max}\sqrt{2\ln(n_v)/n_a(s)}$ be the UCT confidence width for child a after $n_a(s)$ visits within v's sub-problem. By Hoeffding's inequality for sub-Gaussian noise:

$$\mathbb{P}\Big[|\hat{v}_a^{(s)} - V(a)| > c_a(s)\Big] \le \frac{2}{n_v^2},$$

and a union bound over K children and n_v rounds gives high-probability validity of all confidence intervals. On this event, overestimation at each round contributes at most the confidence width $c_a(s)$. Summing over n_v rounds by Cauchy–Schwarz and absorbing logarithmic factors:

$$\mathcal{E}_v \;\le\; 8\sigma_{\max}\sqrt{2n_v \log K}.$$

Depth-D extension via backward induction. Label depth levels 0 (root) to $D-1$ (parents of leaves). At each depth d, let $\mathcal{V}_d$ be the set of internal nodes; each node $v \in \mathcal{V}_d$ is visited n_v times in total. The total visits at depth d sum to T: $\sum_{v\in\mathcal{V}_d} n_v = T$.

Applying the single-node bound at every node at depth d and using Cauchy–Schwarz ($\sum_v \sqrt{n_v} \le \sqrt{|\mathcal{V}_d|}\cdot\sqrt{T}$, since $|\mathcal{V}_d| = K^d$):

$$\sum_{v\in\mathcal{V}_d} \mathcal{E}_v \;\le\; \sum_{v\in\mathcal{V}_d} 8\sigma_{\max}\sqrt{2n_v \log K} \;\le\; 8\sigma_{\max}\sqrt{2\log K}\cdot K^{d/2}\cdot\sqrt{T}.$$

Summing over all D levels and recognising the geometric series:

$$\sum_{t=1}^{T}\mathbb{E}\Big[\hat{v}_{s_t^*} - V(\hat{s}_t^*)\Big] \;\le\; \sum_{d=0}^{D-1} 8\sigma_{\max}\sqrt{2\log K}\cdot K^{d/2}\cdot\sqrt{T} \;=\; 8\sigma_{\max}\sqrt{2T\log K}\cdot\Phi(K,D). \qquad \square$$

$\square$

Remark A.4 (Depth–constant tradeoff). For K=2, D=12: $\Phi(2,12) = \sum_{d=0}^{11} 2^{d/2} \le 153$, so the proof gives $C_1 \le 8\cdot 153\cdot\sigma_{\max}\sqrt{2} \approx 1224\,\sigma_{\max}\sqrt{2}$. This constant is large but fixed (independent of T), confirming the $O(\sqrt{T\log K})$ rate. It is a Cauchy–Schwarz upper bound and typically far exceeds the empirical value: the observed slope 0.635 at T=400 (Figure 1) and the actual regret values (≈ 31 at T=400) are consistent with an effective constant in the range 1–5 $\sigma_{\max}\sqrt{2}$ for the paper's parameters. The gap is expected: the Cauchy–Schwarz step is tight only when all K^d nodes at depth d are visited equally, which does not occur in practice under guided UCT exploration.

Proof of Theorem 5.1. Decompose regret at each step t as

$$V(s_t^*) - V(\hat{s}_t^*) = \underbrace{V(s_t^*) - \hat{v}_{s_t^*}}_{\text{(I) estimation error}} + \underbrace{\hat{v}_{s_t^*} - V(\hat{s}_t^*)}_{\text{(II) selection error}}. \tag{1}$$

Bounding term (II): selection error. Summing over t and applying Lemma A.3:

$$\sum_{t=1}^{T}\mathbb{E}\Big[\hat{v}_{s_t^*} - V(\hat{s}_t^*)\Big] \;\le\; C_1\sqrt{T\log K}, \qquad C_1 = 8\sigma_{\max}\sqrt{2}\,\Phi(K,D).$$

Bounding term (I): estimation error. Term (I) is the *signed* quantity $V(s_t^*) - \hat{v}_{s_t^*}$, not its absolute value. Under the zero-mean noise assumption (Assumption 3.6), the two cases are:

Case $b_t = 1$ (verifier called): $\hat{v}_{s_t^*} = \hat{V}(s_t^*) = V(s_t^*) + \varepsilon_{s_t^*}$, so $\mathbb{E}\Big[V(s_t^*) - \hat{v}_{s_t^*}\Big] = \mathbb{E}[-\varepsilon_{s_t^*}] = 0$.

Case $b_t = 0$ (proxy used): $\hat{v}_{s_t^*} = \hat{V}(s_{\text{anc}}) = V(s_{\text{anc}}) + \varepsilon_{s_{\text{anc}}}$, so

$$\mathbb{E}\Big[V(s_t^*) - \hat{v}_{s_t^*}\Big] = V(s_t^*) - V(s_{\text{anc}}) + \mathbb{E}[-\varepsilon_{s_{\text{anc}}}] = V(s_t^*) - V(s_{\text{anc}}) \;\le\; L\cdot|d(s_t^*) - d(s_{\text{anc}})| \;\le\; \tau_t,$$

where the last inequality uses the condition $b_t = 0$ (i.e. $L\cdot|d(s_t) - d(s_{\text{anc}})| \le \tau_t$) and Assumption 3.4.

Summing over all steps:

$$\begin{aligned}\sum_{t=1}^{T}\mathbb{E}\Big[V(s_t^*) - \hat{v}_{s_t^*}\Big] &= \sum_{t:\,b_t=1} 0 + \sum_{t:\,b_t=0}\mathbb{E}\Big[V(s_t^*) - \hat{v}_{s_t^*}\Big] \\ &\le \sum_{t:\,b_t=0}\tau_t \;\le\; \gamma\sum_{t=1}^{T} t^{-1/2} \;\le\; 2\gamma\sqrt{T}.\end{aligned} \tag{2}$$

The zero-mean property eliminates the $\sigma_{\max}$ contribution from direct-call steps; for proxy steps, the ancestor's noise cancels in expectation.

Combining. Adding term (II) (Lemma A.3) and term (I) (equation (2)), using the shorthand $C_1 = C_1(K, D) = 8\sigma_{\max}\sqrt{2}\,\Phi(K, D)$:

$$\mathbb{E}[R(T)] \;\le\; \underbrace{C_1\sqrt{T\log K}}_{\text{(II) selection}} + \underbrace{2\gamma\sqrt{T}}_{\text{(I) estimation}} .$$

Optimising γ. Setting

$$\gamma^* = \tfrac{C_1}{2}\sqrt{\log K} = 4\sigma_{\max}\sqrt{2\log K}\cdot\Phi(K, D)$$

gives $2\gamma^*\sqrt{T} = C_1\sqrt{T\log K}$, so both terms are equal and

$$\mathbb{E}[R(T)] \;\le\; 2C_1\sqrt{T\log K} \;=\; C(K, D)\sqrt{T\log K}, \quad C(K, D) = 2C_1 = 16\sigma_{\max}\sqrt{2}\,\Phi(K, D). \qquad \square$$

$\square$

A.2 Proof of Theorem 5.3

We prove the bound in two parts corresponding to the exploration and exploitation regimes identified in Remark 5.7.

Lemma A.5 (Single-verification property)**.** *Algorithm 1 invokes the verifier on any node $s \in \mathcal{T}$ at most once.*

Proof. Line 11 of Algorithm 1 stores the observation $(s', \hat{v}(s'))$ in the score memory $\mathcal{M}$ immediately after calling the verifier. At every subsequent visit to s', line 8 finds $s' \in \mathcal{M}$, so $s_{\text{anc}} = s'$ and $\delta = L \cdot 0 = 0 < \tau_t$ for any $\tau_t > 0$; the else branch is taken and the verifier is not called. (In the implementation this is enforced by the `verifier_called` flag on each node.) $\square$

Proof of Theorem 5.3. **Part (a): Single-verification ceiling.** By Lemma A.5, $\mathcal{B}(\pi) \le |\mathcal{T}| \le N_{\mathcal{T}}$, where $N_{\mathcal{T}} = K(K^D - 1)/(K - 1) = K + K^2 + \cdots + K^D$ is the number of non-root nodes (verifier calls can occur at depths 1 through D only, since the root has no ancestor from which to compute a depth gap). Since K and D are fixed problem parameters, $N_{\mathcal{T}}$ is a constant independent of T, and the call fraction $\mathcal{B}/T \le N_{\mathcal{T}}/T \to 0$ as $T \to \infty$.

Part (b): Threshold-triggered refinement. A *threshold-triggered* call requires $s_{\text{anc}} \neq \emptyset$ (re-visit) and $\delta_t = L \cdot |d(s_t) - d(s_{\text{anc}})| > \tau_t$. Since depths lie in $\{0, \ldots, D\}$, we have $\delta_t \le LD$ always. When $\tau_t \ge LD$—equivalently, $t \le (\gamma/(LD))^2 =: T_{\text{free}}$—the threshold dominates every possible depth gap, so no threshold-triggered call occurs. Verifier calls during $t \le T_{\text{free}}$ are exclusively first-visit calls ($s_{\text{anc}} = \emptyset$), which are already bounded in total by $N_{\mathcal{T}}$ via Part (a). The number of threshold-triggered calls is at most $\max(0, T - T_{\text{free}})$; together with Part (a) the total satisfies $\mathcal{B}(\pi) \le N_{\mathcal{T}}$.

Part (c): Savings fraction. Immediate from part (a): $\mathcal{B}(\pi)/T \le N_{\mathcal{T}}/T \to 0$.

Empirical correspondence. For the paper's parameters (K=2, D=12, $N_{\mathcal{T}}$=8,190, γ=1.0, L=0.35): $T_{\text{free}} = (1.0/(0.35 \times 12))^2 \approx 0.06$, so the call-free phase is negligible. In the experimental range $T \le 3200$ (exploration regime, $T \ll N_{\mathcal{T}}$), calls grow approximately as $1.9\sqrt{T}\log T$ empirically. The formal bound $N_{\mathcal{T}} = 8{,}190$ is not tight here; it becomes tight in the exploitation regime ($T \gg N_{\mathcal{T}}$). A tighter bound matching the empirical rate would require analysing UCT's traversal concentration, which we leave to future work. $\square$

A.3 Proof of Theorem 5.5

Proof. We bound *cumulative* regret $R(T) = \sum_{t=1}^{T}[V(s_t^*) - V(\hat{s}_t^*)]$ by lower-bounding the per-step error probability at every step t, not merely at the final step.

Hard instance family. Let $\mathcal{T}$ consist of root s_0 with two children s_+ and s_-. Choose $\Delta = \sigma/(2\sqrt{T})$ and let $\theta \in \{+1, -1\}$ be chosen uniformly at random, defining:

$$\mathcal{I}^{(+)}: \quad V(s_+) = \tfrac{1}{2} + \Delta,\ V(s_-) = \tfrac{1}{2} - \Delta,$$
$$\mathcal{I}^{(-)}: \quad V(s_+) = \tfrac{1}{2} - \Delta,\ V(s_-) = \tfrac{1}{2} + \Delta.$$

Verifier observations satisfy $\hat{V}(s) = V(s) + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$. For $t \geq 2$ (after both nodes have been visited at least once), $s_t^* = s_+$ under $\mathcal{I}^{(+)}$ and $s_t^* = s_-$ under $\mathcal{I}^{(-)}$.

Per-step regret via total variation. Denote by $\mathcal{H}_t$ the algorithm's complete history up to step t and let $n_t = \sum_{i=1}^{t} b_i$ be the total verifier calls through step t. Each call contributes KL divergence $\mathrm{KL}(\mathcal{N}(\Delta, \sigma^2) \| \mathcal{N}(-\Delta, \sigma^2)) = 2\Delta^2/\sigma^2$ between the two instances. By the chain rule and Pinsker's inequality:

$$\mathrm{TV}\Big(\mathcal{H}_t^{(+)}, \mathcal{H}_t^{(-)}\Big) \ \leq\ \sqrt{\tfrac{1}{2}\, n_t \cdot \tfrac{2\Delta^2}{\sigma^2}} \ =\ \Delta\sqrt{\tfrac{n_t}{\sigma^2}}.$$

Since $\hat{s}_t^*$ is a measurable function of $\mathcal{H}_t$, the data-processing inequality and Le Cam's two-point method give:

$$\mathbb{P}[\hat{s}_t^* \neq s_t^*] \ \geq\ \tfrac{1}{2}\Big(1 - \mathrm{TV}\Big(\mathcal{H}_t^{(+)}, \mathcal{H}_t^{(-)}\Big)\Big) \ \geq\ \tfrac{1}{2}\left(1 - \Delta\sqrt{n_t/\sigma^2}\right).$$

When $\hat{s}_t^* \neq s_t^*$, the per-step regret is $V(s_t^*) - V(\hat{s}_t^*) = 2\Delta$. Hence:

$$\mathbb{E}[r_t] \ =\ 2\Delta\, \mathbb{P}[\hat{s}_t^* \neq s_t^*] \ \geq\ \Delta\left(1 - \Delta\sqrt{n_t/\sigma^2}\right).$$

Summing over all steps. Let $B_T = n_T = \sum_{t=1}^{T} b_t$ be the total verifier calls. Since $n_t \leq B_T$ for all t:

$$\mathbb{E}[R(T)] = \sum_{t=2}^{T} \mathbb{E}[r_t] \ \geq\ (T-1)\Delta\left(1 - \Delta\sqrt{B_T/\sigma^2}\right). \tag{3}$$

Setting Δ and concluding. Substituting $\Delta = \sigma/(2\sqrt{T})$:

$$\Delta\sqrt{B_T/\sigma^2} = \frac{\sqrt{B_T}}{2\sqrt{T}} \ \leq\ \frac{1}{2},$$

where the last inequality holds because $B_T \leq T$ (at most one call per step). Inserting into (3):

$$\mathbb{E}[R(T)] \ \geq\ (T-1) \cdot \frac{\sigma}{2\sqrt{T}} \cdot \left(1 - \tfrac{1}{2}\right) \ =\ \frac{\sigma(T-1)}{4\sqrt{T}} \ \geq\ \frac{\sigma}{8}\sqrt{T} \qquad \text{for all } T \geq 2.$$

This bound holds for *every* policy π on this instance family, regardless of the number of verifier calls made. Hence $\mathbb{E}[R(T)] \geq \frac{\sigma}{8}\sqrt{T}$. □

B Hyperparameter Sensitivity

Table 7 reports the sensitivity of DVA-MCTS to the threshold exponent α with $\gamma = 1.0$ fixed, from the ablation experiment (Section 6.9, 50 runs, $T = 400$, $L = 0.35$, $\sigma = 0.05$).

Table 7: Ablation over threshold exponent α at T=400, γ=1.0 (50 runs; all numbers from `results/ablation_threshold.json`).

α	Regret $R(400)$	Verifier Calls	Call Fraction
0.25	27.87	371.9	93.0%
0.50	18.98	382.2	95.5%
0.75	**17.82**	382.7	95.7%
1.00	19.99	387.0	96.7%

The empirically optimal exponent at T=400 is α=0.75 (lowest regret). The theoretically recommended α=0.5 is within 6.5% of optimal regret. As discussed in Section 6.9, verification rates are uniformly high at T=400 because the tree remains largely unexplored; the regime where α most strongly differentiates call patterns is large B ($\geq$800).

C Notation Summary

Table 8: Summary of notation.

Symbol	Description
$\mathcal{T}$	Search tree
$\mathcal{S}, \mathcal{A}$	State and action spaces
K, D	Branching factor and depth
$V(s)$	True verifier score at state s
$\hat{V}(s)$	Noisy verifier observation at s
$\hat{v}_t$	Algorithm's estimate at step t
$b_t \in \{0, 1\}$	Verifier call indicator at step t
τ_t	Budget-sensitive threshold at step t
γ	Threshold constant
α	Threshold decay exponent
L	Lipschitz constant of V
σ	Sub-Gaussian noise parameter
T	Total search budget
$R(T)$	Cumulative regret
$\mathcal{B}(\pi)$	Total verifier calls under policy π
s_t^*	Oracle's best state through step t
$\hat{s}_t^*$	Algorithm's best estimated state through step t
s_{anc}	Deepest ancestor with stored verifier observation
$\mathcal{M}$	Score memory (set of $(s, \hat{V}(s))$ pairs)